0 1000 2000

0

500

1000

1500

2000

2500 Time [s]

0 1000 2000

0

10

20

30

40

Time [min]

0 1000 2000

0.0

0.1

0.2

0.3

0.4

0.5

0.6

0.7 Time [h]

0 1000 2000

0.0

0.1

0.2

0.3

0.4

0.5

0.6

0.7 Discharge capacity [A.h]

0 1000 2000

0.0

0.1

0.2

0.3

0.4

0.5

0.6

0.7 Throughput capacity [A.h]

0 1000 2000

5000

7500

10000

12500

15000

17500

20000

Average negative particle concentration [mol.m-3]

0 1000 2000

5000

7500

10000

12500

15000

17500

20000

X-averaged negative particle surface concentration [mol.m-3]

0 1000 2000

5000

7500

10000

12500

15000

17500

20000

Minimum negative particle concentration [mol.m-3]

0 1000 2000

8000

10000

12000

14000

16000

18000

20000

Maximum negative particle concentration [mol.m-3]

0 1000 2000

5000

7500

10000

12500

15000

17500

20000

Minimum negative particle Minimum negative particle surface concentration [mol.m-3]

0 1000 2000

5000

7500

10000

12500

15000

17500

20000

Maximum negative particle surface concentration [mol.m-3]

0 1000 2000

0.2

0.3

0.4

0.5

0.6

0.7

0.8

Average negative particle concentration

0 1000 2000

0.2

0.3

0.4

0.5

0.6

0.7

0.8

X-averaged negative particle surface concentration

0 1000 2000

0.2

0.3

0.4

0.5

0.6

0.7

0.8

Minimum negative particle concentration

0 1000 2000

0.3

0.4

0.5

0.6

0.7

0.8

Maximum negative particle concentration

0 1000 2000

0.2

0.3

0.4

0.5

0.6

0.7

0.8

Minimum negative particle surface concentration

0 1000 2000

0.2

0.3

0.4

0.5

0.6

0.7

0.8

Maximum negative particle surface concentration

0 1000 2000

0.2

0.3

0.4

0.5

0.6

0.7

0.8

Average negative particle stoichiometry

0 1000 2000

0.2

0.3

0.4

0.5

0.6

0.7

0.8

X-averaged negative particle surface stoichiometry

0 1000 2000

0.2

0.3

0.4

0.5

0.6

0.7

0.8

Minimum negative particle stoichiometry

0 1000 2000

0.3

0.4

0.5

0.6

0.7

0.8

Maximum negative particle stoichiometry

0 1000 2000

0.2

0.3

0.4

0.5

0.6

0.7

0.8

Minimum negative particle surface stoichiometry

0 1000 2000

0.2

0.3

0.4

0.5

0.6

0.7

0.8

Maximum negative particle surface stoichiometry

0 1000 2000

0.2

0.3

0.4

0.5

0.6

0.7

0.8

X-averaged negative electrode extent of lithiation

0 1000 2000

30000

32500

35000

37500

40000

42500

45000

47500

Average positive particle concentration [mol.m-3]

0 1000 2000

30000

35000

40000

45000

X-averaged positive particle surface concentration [mol.m-3]

0 1000 2000

30000

32500

35000

37500

40000

42500

45000

47500

Minimum positive particle concentration [mol.m-3]

0 1000 2000

30000

35000

40000

45000

Maximum positive particle concentration [mol.m-3]

0 1000 2000

30000

35000

40000

45000

Minimum positive particle Minimum positive particle surface concentration [mol.m-3]

0 1000 2000

30000

35000

40000

45000

Maximum positive particle surface concentration [mol.m-3]

0 1000 2000

0.60

0.65

0.70

0.75

0.80

0.85

0.90

0.95 Average positive particle concentration

0 1000 2000

0.60

0.65

0.70

0.75

0.80

0.85

0.90

0.95

X-averaged positive particle surface concentration

0 1000 2000

0.60

0.65

0.70

0.75

0.80

0.85

0.90

0.95 Minimum positive particle concentration

0 1000 2000

0.60

0.65

0.70

0.75

0.80

0.85

0.90

0.95

Maximum positive particle concentration

0 1000 2000

0.60

0.65

0.70

0.75

0.80

0.85

0.90

0.95

Minimum positive particle surface concentration

0 1000 2000

0.60

0.65

0.70

0.75

0.80

0.85

0.90

0.95

Maximum positive particle surface concentration

0 1000 2000

0.60

0.65

0.70

0.75

0.80

0.85

0.90

0.95 Average positive particle stoichiometry

0 1000 2000

0.60

0.65

0.70

0.75

0.80

0.85

0.90

0.95

X-averaged positive particle surface stoichiometry

0 1000 2000

0.60

0.65

0.70

0.75

0.80

0.85

0.90

0.95 Minimum positive particle stoichiometry

0 1000 2000

0.60

0.65

0.70

0.75

0.80

0.85

0.90

0.95

Maximum positive particle stoichiometry

0 1000 2000

0.60

0.65

0.70

0.75

0.80

0.85

0.90

0.95

Minimum positive particle surface stoichiometry

0 1000 2000

0.60

0.65

0.70

0.75

0.80

0.85

0.90

0.95

Maximum positive particle surface stoichiometry

0 1000 2000

0.60

0.65

0.70

0.75

0.80

0.85

0.90

0.95X-averaged positive electrode extent of lithiation

0 1000 2000

0.2

0.3

0.4

0.5

0.6

0.7

0.8

Negative electrode stoichiometry

0 1000 2000

0.2

0.3

0.4

0.5

0.6

0.7

0.8

Negative electrode volume-averaged concentration

0 1000 2000

5000

7500

10000

12500

15000

17500

20000

Negative electrode volume-averaged concentration [mol.m-3]

0 1000 2000

0.010

0.015

0.020

0.025

0.030

0.035

Total lithium in primary phase in negative electrode [mol]

0 1000 2000

0.60

0.65

0.70

0.75

0.80

0.85

0.90

0.95 Positive electrode stoichiometry

0 1000 2000

0.60

0.65

0.70

0.75

0.80

0.85

0.90

0.95Positive electrode volume-averaged concentration

0 1000 2000

30000

32500

35000

37500

40000

42500

45000

47500

Positive electrode volume-averaged concentration [mol.m-3]

0 1000 2000

0.045

0.050

0.055

0.060

0.065

0Total lithium in primary phase in positive electrode [mol].070

0 1000 2000

0.18

0.20

0.22

0.24

0.26

Negative electrode equilibrium open-circuit potential [V]

0 1000 2000

0.18

0.20

0.22

0.24

0.26

X-averaged negative electrode equilibrium open-circuit potential [V]

0 1000 2000

0.18

0.20

0.22

0.24

0.26

X-averaged negative electrode open-circuit potential [V]

0 1000 2000

0.18

0.19

0.20

0.21

0.22

0.23

0.24Negative electrode bulk open-circuit potential [V]

0 1000 2000

0.000

0.005

0.010

0.015

0.020

0.025Negative particle concentration overpotential [V]

0 1000 2000

− 2.5

− 2.0

− 1.5

− 1.0

− 0.5

0.0

×10−5

Negative electrode entropic change [V.K-1]

0 1000 2000

− 2.5

− 2.0

− 1.5

− 1.0

− 0.5

0.0

×10−5

X-averaged negative electrode entropic change [V.K-1]

0 1000 2000

3.5

3.6

3.7

3.8

3.9

4.0

Positive electrode equilibrium open-circuit potential [V]

0 1000 2000

3.5

3.6

3.7

3.8

3.9

4.0

X-averaged positive electrode equilibrium open-circuit potential [V]

0 1000 2000

3.5

3.6

3.7

3.8

3.9

4.0

X-averaged positive electrode open-circuit potential [V]

0 1000 2000

3.75

3.80

3.85

3.90

3.95

4.00

Positive electrode bulk open-circuit potential [V]

0 1000 2000

− 0.25

− 0.20

− 0.15

− 0.10

− 0.05

0.00

Positive particle concentration overpotential [V]

0 1000 2000

− 0.0007

− 0.0006

− 0.0005

− 0.0004

− 0.0003

− 0.0002

− 0.0001

0.0000

Positive electrode entropic change [V.K-1]

0 1000 2000

− 0.0007

− 0.0006

− 0.0005

− 0.0004

− 0.0003

− 0.0002

− 0.0001

0.0000

X-averaged positive electrode entropic change [V.K-1]

0 1000 2000

6.0

6.5

7.0

7.5

8.0

X-averaged negative electrode exchange current density [A.m-2]

0 1000 2000

0.0065

0.0070

0.0075

0.0080

0.0085

X-averaged negative electrode reaction overpotential [V]

0 1000 2000

0.18

0.20

0.22

0.24

0.26

X-averaged negative electrode surface potential difference [V]

0 1000 2000

0.20

0.25

0.30

0.35

0.40

0.45

X-averaged positive electrode exchange current density [A.m-2]

0 1000 2000

− 0.13

− 0.12

− 0.11

− 0.10

− 0.09

X-averaged positive electrode reaction overpotential [V]

0 1000 2000

3.4

3.5

3.6

3.7

3.8

3.9

X-averaged positive electrode surface potential difference [V]

0 1000 2000

− 0.26

− 0.24

− 0.22

− 0.20

− 0.18 X-averaged electrolyte potential [V]

0 1000 2000

− 0.26

− 0.24

− 0.22

− 0.20

− 0.18

X-averaged negative electrolyte potential [V]

0 1000 2000

− 0.26

− 0.24

− 0.22

− 0.20

− 0.18

X-averaged separator electrolyte potential [V]

0 1000 2000

− 0.26

− 0.24

− 0.22

− 0.20

− 0.18

X-averaged positive electrolyte potential [V]

0 1000 2000

0.18

0.20

0.22

0.24

0.26

Negative electrode surface potential difference at separator interface [V]

0 1000 2000

3.1

3.2

3.3

3.4

3.5

3.6

3.7

X-averaged positive electrode potential [V]

0 1000 2000

3.1

3.2

3.3

3.4

3.5

3.6

3.7

Positive current collector potential [V]

0 1000 2000

3.1

3.2

3.3

3.4

3.5

3.6

3.7

Local voltage [V]

0 1000 2000

3.1

3.2

3.3

3.4

3.5

3.6

3.7

Voltage expression [V]

0 1000 2000

3.1

3.2

3.3

3.4

3.5

3.6

3.7

Terminal voltage [V]

0 1000 2000

3.1

3.2

3.3

3.4

3.5

3.6

3.7

Voltage [V]

0 1000 2000

3.4

3.5

3.6

3.7

3.8

3.9

Positive electrode surface potential difference at separator interface [V]

0 1000 2000

3.3

3.4

3.5

3.6

3.7

3.8

Surface open-circuit voltage [V]

0 1000 2000

3.50

3.55

3.60

3.65

3.70

3.75

3.80

3.85 Bulk open-circuit voltage [V]

0 1000 2000

− 0.25

− 0.20

− 0.15

− 0.10

− 0.05

0.00

Particle concentration overpotential [V]

0 1000 2000

− 0.13

− 0.12

− 0.11

− 0.10

− 0.09

X-averaged reaction overpotential [V]

0 1000 2000

3.50

3.55

3.60

3.65

3.70

3.75

3.80

3.85 Battery open-circuit voltage [V]

0 1000 2000

0.18

0.19

0.20

0.21

0.22

0.23

0.24 Battery negative electrode bulk open-circuit potential [V]

0 1000 2000

3.75

3.80

3.85

3.90

3.95

4.00

Battery positive electrode bulk open-circuit potential [V]

0 1000 2000

− 0.25

− 0.20

− 0.15

− 0.10

− 0.05

0.00

Battery particle concentration overpotential [V]

0 1000 2000

0.000

0.005

0.010

0.015

0.020

0Battery negative particle concentration overpotential [V].025

0 1000 2000

− 0.25

− 0.20

− 0.15

− 0.10

− 0.05

0.00

Battery positive particle concentration overpotential [V]

0 1000 2000

− 0.13

− 0.12

− 0.11

− 0.10

− 0.09

X-averaged battery reaction overpotential [V]

0 1000 2000

0.0065

0.0070

0.0075

0.0080

0.0085

X-averaged battery negative reaction overpotential [V]

0 1000 2000

− 0.13

− 0.12

− 0.11

− 0.10

− 0.09

X-averaged battery positive reaction overpotential [V]

0 1000 2000

3.1

3.2

3.3

3.4

3.5

3.6

3.7

Battery voltage [V]

0 1000 2000

0.10

0.15

0.20

0.25

0.30

0.35

0.40

Local ECM resistance [Ohm]

0 1000 2000

3.1

3.2

3.3

3.4

3.5

3.6

3.7

Terminal power [W]

0 1000 2000

3.1

3.2

3.3

3.4

3.5

3.6

3.7

Power [W]

0 1000 2000

3.1

3.2

3.3

3.4

3.5

3.6

3.7

Resistance [Ohm]

0 1000 2000

0.010

0.015

0.020

0.025

0.030

0.035

Total lithium in negative electrode [mol]

0 1000 2000

0.045

0.050

0.055

0.060

0.065

0.070 Total lithium in positive electrode [mol]

0 1000 2000

0.0

0.5

1.0

1.5

2.0

×10−12
LLI [

0 1000 2000

0.0

0.5

1.0

1.5

2.0

×10−12
Loss of lithium inventory [

0 1000 2000

0.0

0.5

1.0

1.5

2.0

×10−12
Loss of lithium inventory, including electrolyte [

0 1000 2000

4.25

4.50

4.75

5.00

5.25

5.50

5.75

×10−13 + 7.999320536 × 10−2
Total lithium [mol]

0 1000 2000

4.4

4.6

4.8

5.0

5.2

5.4

5.6

5.8

×10−13 + 7.758269036 × 10−2
Total lithium in particles [mol]

0 1000 2000

2

3

4

5

6

×10−12 + 2.143936385
Total lithium capacity [A.h]

0 1000 2000

9

10

11

12

×10−12 + 2.079331013
Total lithium capacity in particles [A.h]

0 1000 2000

0.00

0.25

0.50

0.75

1.00

1.25

1.50

×10−15
Total lithium lost [mol]

0 1000 2000

0.00

0.25

0.50

0.75

1.00

1.25

1.50

×10−15
Total lithium lost from particles [mol]

0.0 0.5 1.0

0.0

0.2

0.4

0.6

0.8

1.0

0.0 0.5 1.0

0.0

0.2

0.4

0.6

0.8

1.0

0.0 0.5 1.0

0.0

0.2

0.4

0.6

0.8

1.0

0.0 0.5 1.0

0.0

0.2

0.4

0.6

0.8

1.0

0.0 0.5 1.0

0.0

0.2

0.4

0.6

0.8

1.0

0.0 0.5 1.0

0.0

0.2

0.4

0.6

0.8

1.0

0.0 0.5 1.0

0.0

0.2

0.4

0.6

0.8

1.0

0.0 0.5 1.0

0.0

0.2

0.4

0.6

0.8

1.0

0.0 0.5 1.0

0.0

0.2

0.4

0.6

0.8

1.0

0.0 0.5 1.0

0.0

0.2

0.4

0.6

0.8

1.0

0.0 0.5 1.0

0.0

0.2

0.4

0.6

0.8

1.0

0.0 0.5 1.0

0.0

0.2

0.4

0.6

0.8

1.0

0.0 0.5 1.0

0.0

0.2

0.4

0.6

0.8

1.0

0.0 0.5 1.0

0.0

0.2

0.4

0.6

0.8

1.0

0.0 0.5 1.0

0.0

0.2

0.4

0.6

0.8

1.0

0.0 0.5 1.0

0.0

0.2

0.4

0.6

0.8

1.0

0.0 0.5 1.0

0.0

0.2

0.4

0.6

0.8

1.0

0.0 0.5 1.0

0.0

0.2

0.4

0.6

0.8

1.0

0.0 0.5 1.0

0.0

0.2

0.4

0.6

0.8

1.0

0.0 0.5 1.0

0.0

0.2

0.4

0.6

0.8

1.0

0.0 0.5 1.0

0.0

0.2

0.4

0.6

0.8

1.0

0.0 0.5 1.0

0.0

0.2

0.4

0.6

0.8

1.0

0.0 0.5 1.0

0.0

0.2

0.4

0.6

0.8

1.0

0.0 0.5 1.0

0.0

0.2

0.4

0.6

0.8

1.0

0.0 0.5 1.0

0.0

0.2

0.4

0.6

0.8

1.0

0.0 0.5 1.0

0.0

0.2

0.4

0.6

0.8

1.0